Google DSA Technical Interview Topic-Wise Question Bank

Complete Curated Collection Across All Core Data Structures & Algorithms

GOOGLE L3 / L4 / L5 | ALL DSA TOPICS | LEETCODE INDEXED

1. Arrays, Matrix & Two Pointers

Question Name	Difficulty	Primary Pattern	Core Technique & Google Context
Trapping Rain Water (LC 42)	Hard	`Two Pointers`	Tracks left/right maximum elevations to maintain O(1) space.
First Missing Positive (LC 41)	Hard	`Cyclic Sort`	In-place array index hashing to achieve O(N) time and O(1) space.
3Sum (LC 15)	Medium	`Two Pointers`	Sort + left/right pointers; requires careful duplicate elimination.
Subarray Sum Equals K (LC 560)	Medium	`Prefix Sum + Map`	Tracks prefix sum frequencies for contiguous subarray matching.
Merge Intervals (LC 56)	Medium	`Sorting Intervals`	Sort by start times and merge overlapping bounds iteratively.

2. Strings & Pattern Matching

Question Name	Difficulty	Primary Pattern	Core Technique & Google Context
Minimum Window Substring (LC 76)	Hard	`Sliding Window`	Frequency map contraction to find minimal valid substring bounds.
Longest Substring No Repeat (LC 3)	Medium	`Sliding Window`	Tracks character index map to jump left pointer on duplicate encounter.
Group Anagrams (LC 49)	Medium	`Hash Map Key`	Categorize strings using tuple character frequency or sorted string keys.
Palindromic Substrings (LC 647)	Medium	`Expand Around Center`	Expands around odd/even centers for $O(N^2)$ time & O(1) space checks.

3. Stack, Queue & Monotonic Structures

Question Name	Difficulty	Primary Pattern	Core Technique & Google Context
Sliding Window Maximum (LC 239)	Hard	`Monotonic Deque`	Maintains indices in decreasing value order for O(1) max queries.
Largest Rectangle in Histogram (LC 84)	Hard	`Monotonic Stack`	Tracks increasing bar heights to calculate rectangle bounds on drop.
Basic Calculator II (LC 227)	Medium	`Stack Expression`	Handles operator precedence (*, / over +, -) using stack aggregation.
Daily Temperatures (LC 739)	Medium	`Monotonic Stack`	Next greater element pattern storing unresolved indices.

4. Linked List

Question Name	Difficulty	Primary Pattern	Core Technique & Google Context
Merge k Sorted Lists (LC 23)	Hard	`Min-Heap / Divide`	Uses a Priority Queue of node heads or divide-and-conquer merges.
Reverse Nodes in k-Group (LC 25)	Hard	`In-Place Pointer`	Reverses linked list chunks of size k while maintaining boundary links.
Copy List with Random Pointer (LC 138)	Medium	`Hash Map / Interleave`	Clones nodes via map or by interleaving original and copied nodes.
LRU Cache (LC 146)	Medium	`DLL + Hash Map`	Doubly Linked List for O(1) eviction + Hash Map for O(1) lookup.

5. Binary Trees & Search Trees

Question Name	Difficulty	Primary Pattern	Core Technique & Google Context
Binary Tree Max Path Sum (LC 124)	Hard	`DFS Recursion`	Calculates max sub-path contribution while updating global path sum.
Serialize & Deserialize Tree (LC 297)	Hard	`Preorder BFS / DFS`	Converts tree to string stream using null markers for reconstruction.
Lowest Common Ancestor (LC 236)	Medium	`Tree Postorder`	Bubbles up target matches from left and right subtrees.
Build Tree Preorder/Inorder (LC 105)	Medium	`Divide & Conquer`	Uses root index in preorder to split inorder array recursively.

6. Graphs & Network Topologies

Question Name	Difficulty	Primary Pattern	Core Technique & Google Context
Shortest Path Grid Obstacles (LC 1293)	Hard	`3D State BFS`	Queue tracks (r, c, remaining_k) to find shortest path layered.
Alien Dictionary (LC 269)	Hard	`Topological Sort`	Builds character dependency DAG and executes Kahn's BFS queue sort.
Course Schedule II (LC 210)	Medium	`Indegree Queue`	Detects cycles and orders prerequisite courses using topological sort.
Number of Islands (LC 200)	Medium	`BFS / DFS / Union-Find`	Sinks connected components or merges adjacent set components.

7. Heap & Priority Queue

Question Name	Difficulty	Primary Pattern	Core Technique & Google Context
Find Median from Data Stream (LC 295)	Hard	`Two Heaps`	Balances Max-Heap (lower half) & Min-Heap (upper half) for O(1) median.
Kth Largest Element (LC 215)	Medium	`Min-Heap / QuickSelect`	Min-Heap of size K or QuickSelect partitioning for O(N) average.
Top K Frequent Elements (LC 347)	Medium	`Bucket Sort / Heap`	Frequency map paired with size K Min-Heap or Bucket Sort array.

8. Trie & Custom System Design

Question Name	Difficulty	Primary Pattern	Core Technique & Google Context
Search Autocomplete System (LC 642)	Hard	`Trie + Top-3 Cache`	Prefix trie node caches top 3 sentences for instant O(1) lookup.
Snapshot Array (LC 1146)	Medium	`Array + Binary Search`	Stores (snap_id, val) list per index; queries via bisect_right.
Implement Trie (LC 208)	Medium	`Prefix Tree`	Tree structure with 26-child array/map and end-of-word boolean.

9. Dynamic Programming & Backtracking

Question Name	Difficulty	Primary Pattern	Core Technique & Google Context
Edit Distance (LC 72)	Hard	`2D Grid DP`	Computes minimal insert/delete/replace operations dynamically.
Word Break II (LC 140)	Hard	`Memoized Backtrack`	DFS with memoization map storing valid sentence combinations.
Coin Change (LC 322)	Medium	`Unbounded Knapsack`	Bottom-up 1D DP table computing minimum coins for each target sum.
Word Search (LC 79)	Medium	`Backtracking DFS`	Grid DFS marking visited cells temporarily during path exploration.